

Passive Drone Detection by Bayesian Fusion of Event-Camera Propeller Sensing and Microphone-Array Acoustic Localization

Author Name
Affiliation
Organization
City, Country
author@email

Abstract—Small unmanned aerial vehicles (UAVs) are difficult to detect with conventional radar and RGB cameras because of their low radar cross-section, small visual footprint, and low acoustic energy against urban background noise. This paper presents NeuraSense, a passive dual-modality detector that fuses a neuromorphic event camera (Prophesee EVK4-HD, IMX636) with a four-microphone acoustic array (ReSpeaker). The event camera exploits the microsecond temporal resolution and >120 dB dynamic range of the sensor to recover the per-pixel blade-pass frequency (BPF) of a rotating propeller, while the microphone array estimates the same BPF and its direction of arrival (DOA) using frequency-weighted GCC-PHAT. A sequential Bayesian filter fuses the two modalities and rejects single-modality false alarms through a harmonic frequency cross-validation step. On recorded event data the visual stage locks a propeller at 99.7 Hz (2990 RPM) with a stable centroid and zero latency drift over 175 s, and the acoustic stage independently confirms a 182–189 Hz blade-pass tone. We report the full processing pipeline, the fusion mathematics, latency budget, and an honest analysis of the cross-modal spatial-calibration limits.

Index Terms—Event camera, neuromorphic vision, drone detection, propeller blade-pass frequency, GCC-PHAT, direction of arrival, Bayesian sensor fusion, acoustic localization.

I. Introduction

Counter-UAV sensing must contend with targets that are small, slow, and often flying below the radar horizon. Passive electro-optical and acoustic approaches are attractive because they emit nothing and are inexpensive, but each modality alone is fragile: cameras fail in low light or clutter, and single-array acoustics is easily masked by ambient noise and suffers from poor angular resolution at long range.

A rotating propeller is, however, a strong periodic signature in both domains simultaneously. Optically, each blade sweep modulates the brightness of the pixels it crosses at the blade-pass frequency $f_{\text{BPF}} = (\text{RPM}/60) \cdot N_{\text{blades}}$. Acoustically, the same blade passing produces a tone at the identical fundamental. This shared, physically-linked frequency is the anchor of the present work: if two independent sensors report the same BPF from the same direction at the same time, the joint probability of a false alarm collapses.

We use an event camera rather than a frame camera because propeller detection is fundamentally a high-speed temporal task. A frame camera at 30–60 fps aliases blade-pass frequencies of hundreds of Hz; the EVK4-HD timestamps each pixel-level brightness change to $\sim 1 \mu\text{s}$, so periodic motion up to the kHz range is recoverable directly from the event stream without motion blur and across a >120 dB dynamic range.

Contributions.

- 1) A latency-stable per-pixel BPF detector built on the Metavision FrequencyMapAsyncAlgorithm, redesigned from a failed grid-FFT baseline into a connected-component + temporal-tracking pipeline (Sec. IV).
- 2) A frequency-weighted, zero-padded GCC-PHAT DOA estimator tuned for the low-frequency propeller band on a compact 64 mm-baseline array (Sec. V).
- 3) A sequential Bayesian fusion filter with harmonic frequency cross-validation and temporal association gating (Sec. VI).
- 4) A measured evaluation and a candid discussion of where the current cross-modal spatial calibration is not yet reliable (Sec. VII–VIII).

II. Related Work

Radar-based counter-UAV systems dominate the literature but struggle with micro-UAVs of low radar cross-section and require active emission. Acoustic UAV detection using microphone arrays and machine learning has been shown to work at ranges of tens to low-hundreds of metres in quiet conditions, but degrades sharply with wind and traffic noise, and small arrays give coarse DOA. RGB-vision detectors (e.g., YOLO-family) are effective when the UAV is large in frame but fail at range and in poor illumination.

Neuromorphic (event) cameras have recently been applied to high-speed vibration and rotation sensing, exploiting their microsecond latency and sparse output [2]. Our work differs by (i) targeting the propeller rather than the airframe, using the blade-pass frequency as an invariant,

TABLE I
Hardware and Compute Components

Component	Device	Key specification
Vision	Prophesee EVK4-HD	1280×720, ~1 μs, >120 dB DR
Audio	ReSpeaker 4-Mic	4 MEMS, 32 mm circular, 10 kHz
Compute	Python 3.9 + SDK 4.6.2	OpenCV, NumPy, PyAudio

and (ii) fusing the event-based BPF estimate with an acoustic estimate of the same physical quantity so that each modality validates the other.

III. System Overview

NeuraSense comprises two passive sensors and a fusion core.

The two pipelines run in separate threads. The visual pipeline processes the event stream on the main thread; an AcousticProcessor daemon thread continuously buffers audio and publishes its latest result. The fusion core reads both, applies a temporal association gate, and performs a sequential Bayesian update.

IV. Visual Pipeline: Event-Based Blade-Pass Detection

A. Per-Pixel Frequency Map

Events are sliced by the EventsIterator into $\Delta t = 500 \mu s$ buckets (2000 Hz effective sampling, Nyquist 1000 Hz). The Metavision FrequencyMapAsyncAlgorithm maintains, for every pixel, the frequency of its periodic brightness modulation, producing a 1280×720 frequency map updated at 10–20 Hz.

B. From Grid-FFT to Connected Components

The initial baseline divided the sensor into a 16×16 grid and ran an FFT on each cell’s event-rate time series. This failed beyond a few centimetres: a 160×90 px cell mixes a handful of propeller pixels with hundreds of noise pixels, diluting the periodic signal below the noise floor, and lowering thresholds merely converted the noise floor into 25–43 false “regions” per frame. The key lesson was that parameter tuning cannot repair a wrong architecture.

The redesigned pipeline is: (1) threshold the frequency map into a binary mask of pixels whose frequency lies in $[f_{\min}, f_{\max}]$; (2) downscale 4× (1280×720 → 320×180) and dilate with an elliptical kernel; (3) connected-component labelling (8-connectivity) to obtain spatial clusters; (4) filter clusters by minimum pixel count (default 3) and frequency coefficient of variation $CV = \sigma_f / \mu_f \leq 0.3$.

C. Temporal Tracking and Confidence

Candidate detections are matched to existing tracks by spatial proximity (≤ 100 px) and frequency similarity ($\leq 30\%$ relative). Track state is smoothed with an exponential moving average, $\alpha = 0.3$. A track’s confidence follows a Bayesian persistence model

$$P(\text{propeller}) = 1 - (1 - p)^n, \quad p \in [0.65, 0.85], \quad (1)$$

where n is the number of consecutive hits. A track is confirmed when $P \geq 0.95$ and $n \geq 2$; tracks unseen for $\text{max_age} = 8$ frames are pruned, removing transient 3-pixel noise clusters within 1–2 s.

D. Latency Elimination

Running clustering synchronously on the full-resolution map at 25 Hz cost ~15 ms per callback (~375 ms per second of budget), causing monotonically increasing display lag. Four fixes removed the drift: 4× downscale before morphology/CC (~10× speedup); update rate 25 → 10 Hz; in-place drawing (no frame.copy(), removing ~67 MB/s allocation); and early exit on empty masks. Net callback cost fell from $\sim 15 \text{ ms} \times 25 \text{ s}^{-1}$ to $\sim 2 \text{ ms} \times 10 \text{ s}^{-1}$, giving zero latency drift over 175 s.

V. Acoustic Pipeline: BPF and Direction of Arrival

A. Blade-Pass Frequency

Each of the four channels is accumulated into a 0.2 s ring buffer (3200 samples at 16 kHz). A Hann-windowed FFT yields per-channel spectra; the dominant peak in $[f_{\min}, f_{\max}]$ is the acoustic BPF, with SNR measured as peak magnitude over the spectral median.

B. Frequency-Weighted GCC-PHAT

DOA is estimated from the two orthogonal 64 mm mic pairs (0–2 and 1–3). For a pair of signals with spectra S_1, S_2 , the generalized cross-correlation with phase transform is weighted toward the target band:

$$R(\tau) = \mathcal{F}^{-1} \left\{ \frac{S_1 S_2^*}{|S_1 S_2^*|} (1 + 4w(f)) \right\}, \quad (2)$$

$$w(f) = \frac{|S_1| + |S_2|}{\max(|S_1| + |S_2|)} \Big|_{f \in [f_{\min}, f_{\max}]}. \quad (3)$$

The inverse transform uses 4× zero-padding for sub-sample TDOA resolution, and a main-peak-to-second-peak confidence metric rejects unreliable windows. The azimuth follows from $\theta = \arcsin(c \hat{\tau} / d)$ with $c = 343 \text{ m/s}$ and $d = 0.064 \text{ m}$. Estimates from the two pairs are averaged when both are confident; the planar array cannot resolve elevation.

VI. Bayesian Sensor Fusion

A. Sequential Update

Let $P(D)$ be the current belief that a drone is present. Given visual evidence V and acoustic evidence A , each modality applies a Bayes update

$$P(D | E) = \frac{P(E | D) P(D)}{P(E | D) P(D) + P(E | \bar{D}) P(\bar{D})}. \quad (4)$$

Likelihoods are sigmoids of SNR, gated by the frequency band: $P(V | D) = \sigma(\text{SNR}_V - 5)$ and $P(A | D) = \sigma(\text{SNR}_A - 4)$. Evidence outside $[f_{\min}, f_{\max}]$ receives deliberately weak likelihoods so out-of-band noise cannot drive detection.

TABLE II
First-Detect Latency Budget

Stage	Latency
Visual	~40–60 ms (≈ 7 periods @ 189 Hz + overhead)
Acoustic	~50–120 ms (FFT window fill + USB buffer)
Fused	~100–200 ms (both modalities agree)

B. Harmonic Cross-Validation

When both modalities are present, a cross-validation term rewards agreement while tolerating that one sensor may see a harmonic:

$$\exists n_v, n_a \in \{1..4\} : \frac{|f_V/n_v - f_A/n_a|}{f_A/n_a} < 0.05 \Rightarrow \text{bonus} = \frac{0.99}{n_v n_a}. \quad (5)$$

A match injects a strong likelihood ratio ($P_f(D) = 0.99 \cdot \text{conf vs. } P_f(\bar{D}) = 0.001$), sharply increasing $P(D)$.

C. Decay and Temporal Association

With no new evidence the belief relaxes toward the prior, $P(D) \leftarrow 0.995 P(D) + 0.005 P_0$ with $P_0 = 10^{-3}$. Cross-modal evidence is fused only if visual and acoustic timestamps fall within an association window (default 0.20 s); otherwise a visual-only update is applied. A drone is declared when $P(D) > 0.8$.

VII. Results

A. Visual Detection

On recorded event data the redesigned pipeline produces a single confirmed propeller, stable at 99.7 Hz (≈ 2990 RPM), with a locked centroid at pixel (430, 22), ~ 100 vibrating pixels, and 270+ consecutive hits over the recording. Transient false positives are pruned within 1–2 s.

B. Acoustic Verification

A separate acoustic verification session reported a dominant blade-pass tone. A representative approved detection had a top frequency of 182.1 Hz with a 62.2 dB peak and strong energy across 152–200 Hz, consistent with a ~ 180 –190 Hz fundamental. Windows lacking a clear in-band peak were correctly rejected.

C. Cross-Modal Spatial Calibration (Negative Result)

We attempted an online linear calibration mapping visual azimuth to acoustic azimuth. Over $n = 101$ samples in 7 bins the fit was $\theta_{ac} = -0.011 \theta_{vis} + 0.37^\circ$ with RMSE 0.70° but $R^2 = 0.14$, and the collector flagged the model as not solid. The acoustic azimuth spanned only $\approx 2.5^\circ$ while the visual azimuth spanned $\approx 56^\circ$: the compact array did not resolve the angular motion the camera saw. Thus the current geometry supports frequency-level and temporal fusion, but not yet reliable spatial (azimuth) fusion.

VIII. Discussion and Limitations

- No hardware clock synchronization. Visual events carry μs timestamps while audio is wall-clock buffered; fusion relies on a soft 0.2 s association gate rather than a shared clock.
- Planar 32 mm array. Elevation is unobservable, and the 64 mm baseline gives coarse azimuth—adequate for gating, insufficient for triangulation.
- Harmonic ambiguity. A 2-blade BPF can alias a 4-blade prop at half frequency; cross-validation tolerates harmonics but cannot always disambiguate blade count.
- Limited ground truth. Evaluation used single recordings; no labelled multi-drone dataset or independent error rates yet.
- Range envelope. Visual range is ~ 15 m for Mavic-class targets; acoustic range of 100–150 m in quiet conditions is unvalidated.

IX. Conclusion and Future Work

We presented a passive, dual-modality drone detector that anchors fusion on the physically-shared propeller blade-pass frequency. The event-based visual stage achieves stable, drift-free BPF estimation, the acoustic stage independently confirms the same tone, and a sequential Bayesian filter with harmonic cross-validation combines them. The honest negative result on azimuth calibration defines the next steps: (i) a hardware or software clock sync between sensors, (ii) a larger-baseline or multi-node array for usable DOA, (iii) a labelled multi-drone dataset with reported ROC performance, and (iv) triangulation across 3–4 nodes for 360° coverage and 3D localization.

Acknowledgment

The authors thank [collaborators / funding].

References

- [1] Prophesee, “Metavision SDK—Frequency and vibration estimation algorithms,” Technical documentation, v4.6.
- [2] G. Gallego et al., “Event-based vision: A survey,” *IEEE Trans. Pattern Anal. Mach. Intell.*, 2022.
- [3] C. Knapp and G. Carter, “The generalized correlation method for estimation of time delay,” *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 24, no. 4, pp. 320–327, 1976.
- [4] Seeed Studio, “ReSpeaker 4-Mic Array—hardware and DOA,” Product documentation.
- [5] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. Cambridge, MA: MIT Press, 2005.
- [6] E. E. Case et al., “Low-cost acoustic array for small UAV detection and tracking,” review of array-based UAV acoustics.